

Synth-Reflect AI: Advanced Synthetic Data Generation & Edge-AI Detection for Night-Time Highway Safety

Executive Summary

To solve the inherent dangers and extreme costs associated with manually collecting night-time data on the National Highways Authority of India (NHAI) network, this project introduces Synth-Reflect AI. We have engineered a highly accurate 3D physics simulation to generate infinite, perfectly labeled synthetic training data. We utilized this data to train a lightweight YOLOv8 Nano Machine Learning model that instantly detects failing retroreflective road signs using standard, low-cost dashcams. By bypassing the dangerous physical data collection phase, Synth-Reflect AI offers a zero-cost data pipeline that empowers edge-computing devices to perform real-time, automated infrastructure auditing across India's rapidly expanding highway network.

1. Problem Context

1.1 The Expansion of the Indian National Highway Network

The infrastructure landscape in India has undergone a massive transformation. As of 2024, India possesses the second-largest road network in the world, with National Highways spanning a total length of 146,195 km. This represents a staggering 60% growth over the last decade, driven by flagship programs such as the Bharatmala Pariyojana. While this expansion drastically improves national connectivity and freight movement, it simultaneously creates an enormous infrastructural footprint that requires constant, rigorous maintenance and safety auditing to protect motorists.

1.2 The Burden of Night-Time Road Accidents

Despite infrastructural improvements, road safety remains a critical national challenge. According to the Ministry of Road Transport and Highways (MoRTH) and the National Crime Records Bureau (NCRB), India recorded over 153,972 road crash fatalities in a single recent year. A disproportionate amount of severe accidents occur during twilight and night hours; NCRB data indicates that the 6:00 PM to 9:00 PM window alone accounts for approximately 19% of all total road accidents. During these hours, driver visibility is severely compromised, making the presence of highly visible, functional retroreflective signage an absolute necessity to prevent lane departures, intersection

collisions, and highway speeding fatalities.

1.3 The Bottleneck in Infrastructure Auditing

Currently, identifying degraded road signs requires physical inspection. Manual inspections are prohibitively expensive, slow, and highly dangerous for workers who must navigate active highways in the dark. Advanced technological alternatives, such as LiDAR scanning vehicles, represent massive capital expenditures that are difficult to scale across 146,000+ kilometers of road. Furthermore, training an AI model to automate this process presents a "cold start" data problem: collecting thousands of real-world images of broken, unreflective highway signs at night requires the very dangerous physical driving we are trying to eliminate.

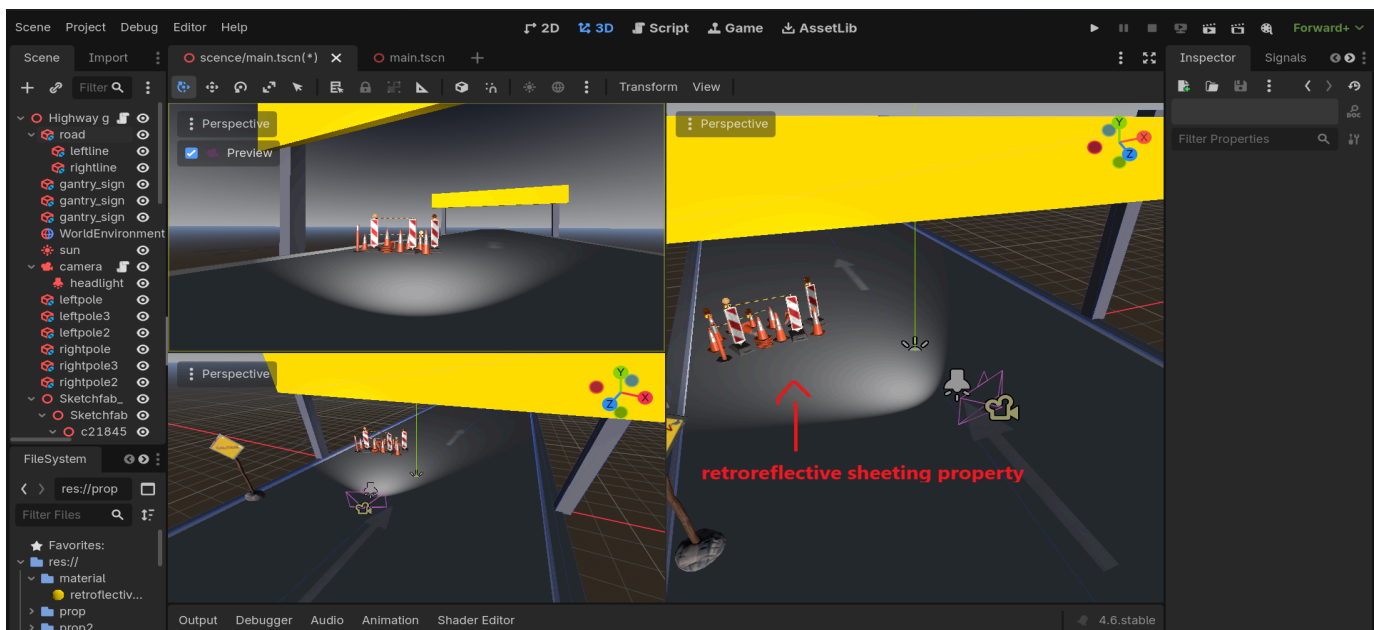
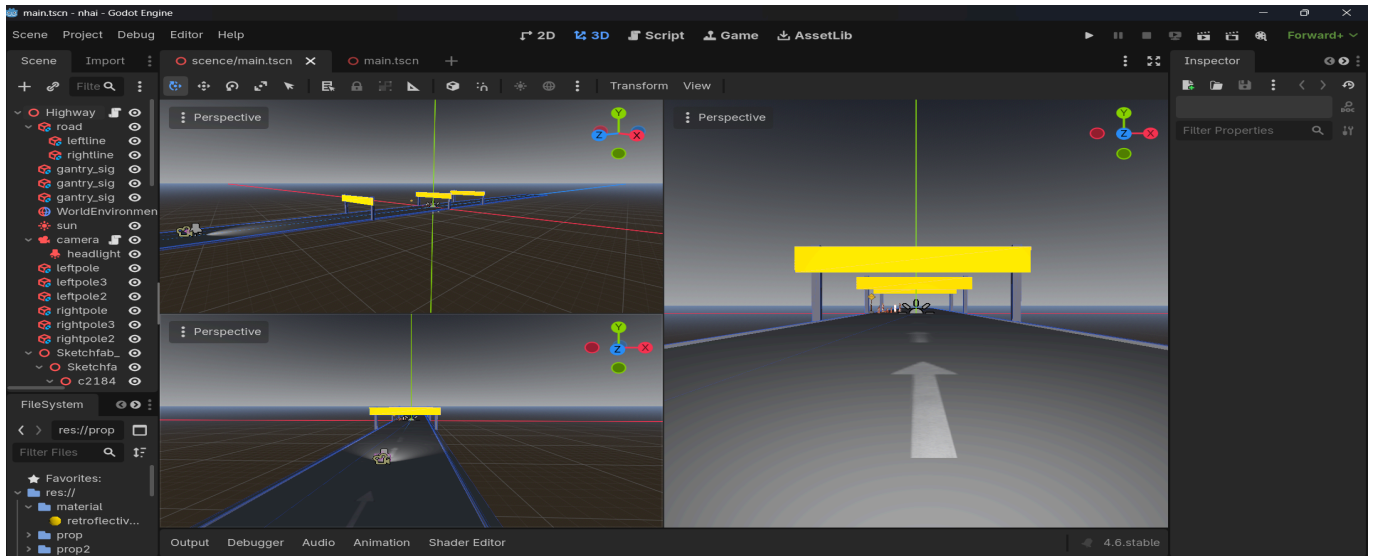
2. The Innovation: Synthetic Data Generation

2.1 Bypassing the Data Bottleneck

The primary innovation of Synth-Reflect AI lies in completely circumventing the physical data collection phase. Recognizing that traditional computer vision models require vast datasets of edge-case scenarios (e.g., shattered signs, mud-covered reflectors, extreme weather), we pivoted to a synthetic generation approach. This guarantees that our model is trained on a perfectly diverse dataset without ever putting a human on a dark highway.

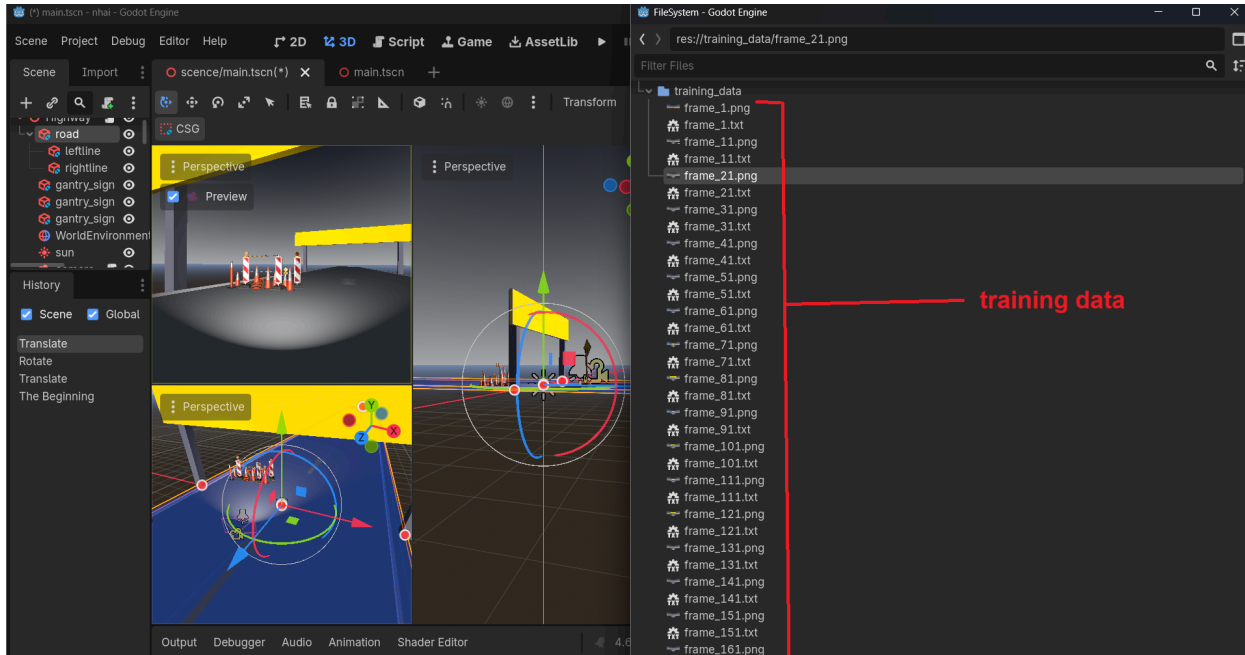
2.2 Godot Engine as a Physics-Accurate Simulation Tool

We utilized the Godot Engine to build a physics-accurate 3D simulation of Indian highways. Within this environment, we programmed specific lighting algorithms to mimic the exact lumen output and beam angles of standard vehicular headlights. The simulation meticulously recreates the material properties of retroreflective sheeting, calculating the angle of incidence and light return to the camera lens. By dynamically altering the condition of the virtual signs (applying digital noise, lowering reflectivity coefficients, and introducing environmental obstructions), the engine produces an infinite array of training scenarios.



2.3 Automated Annotation Pipeline

Manual data labeling is incredibly labor-intensive. Our simulation automates this entirely. As the virtual camera drives down the synthetic highway, the engine continuously outputs frames while simultaneously generating precise bounding-box coordinates for every sign in the viewport. This results in mathematically perfect `.txt` label files, eliminating human error in the annotation process and drastically accelerating the model training timeline.



3. Alignment with IRC:67 Standards

3.1 Code of Practice for Road Signs

Any technological deployment on Indian highways must adhere to the rigorous standards set forth by the Indian Roads Congress (IRC). Specifically, IRC:67 ("Code of Practice for Road Signs") serves as the national benchmark governing the design, placement, retroreflectivity, and uniformity of all traffic control devices. Synth-Reflect AI is specifically tailored to uphold these operational standards.

3.2 Automating Compliance Verification

IRC:67 mandates that retro-reflective sheeting (Class A, B, and C) maintains specific efficiency thresholds over its warranty period (5 to 10 years). The Code requires at least biannual inspections, particularly during night and rainy seasons, to ensure that visibility is not compromised by fading, mud, or vandalism. Synth-Reflect AI automates this exact compliance check. By continually scanning the road via dashcams, the AI flags signs that fail to return the expected luminance footprint, immediately alerting maintenance crews to replace signs that have fallen below the IRC:67 acceptable threshold.

4. Feasibility and Edge Computing Architecture

4.1 Transitioning to Edge Inference

A major flaw in many modern AI propositions is their reliance on constant, high-bandwidth internet connections to transmit video to cloud servers for processing. This is functionally impossible on remote stretches of national highways where 4G/5G connectivity is intermittent. Synth-Reflect AI solves this by relying exclusively on Edge Computing.

4.2 The YOLOv8 Nano Architecture

We consciously selected the Ultralytics YOLOv8 Nano (n) model. YOLO ("You Only Look Once") is the industry standard for real-time object detection. The "Nano" variant is highly compressed and optimized for environments with minimal computational power. This allows the AI to process video frames in real-time directly on the edge device—such as a low-cost onboard computer or the dashcam unit itself. The system only requires internet access momentarily to transmit a tiny text payload (GPS coordinates and sign classification) when a defective sign is found, rather than streaming gigabytes of video data.



```
...
    all      2615      2476      0.987      0.86      0.947      0.834

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
48/50   2.09G    0.6037    0.459     0.7881     7          640: 100%
Class   Images  Instances  Box(P    R          mAP50  mAP50-95): 100%
all     2615    2476      0.992     0.864     0.947    0.834
                                     164/164 2.6it/s 1:04
                                     82/82 2.8it/s 29.8s

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
49/50   2.09G    0.5771    0.4421    0.7841     7          640: 100%
Class   Images  Instances  Box(P    R          mAP50  mAP50-95): 100%
all     2615    2476      0.994     0.861     0.947    0.822
                                     164/164 2.6it/s 1:04
                                     82/82 2.7it/s 30.3s

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
50/50   2.09G    0.5734    0.4365    0.7832     6          640: 100%
Class   Images  Instances  Box(P    R          mAP50  mAP50-95): 100%
all     2615    2476      0.997     0.864     0.95     0.836
                                     164/164 2.6it/s 1:02
                                     82/82 2.7it/s 30.2s

50 epochs completed in 1.336 hours.
Optimizer stripped from /content/nhai_training/night_vision_model/weights/last.pt, 6.2MB
Optimizer stripped from /content/nhai_training/night_vision_model/weights/best.pt, 6.2MB

Validating /content/nhai_training/night_vision_model/weights/best.pt...
ultralytics 8.4.41 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
Class   Images  Instances  Box(P    R          mAP50  mAP50-95): 100%
all     2615    2476      0.999     0.854     0.941    0.839
                                     82/82 2.4it/s 33.9s

Speed: 0.2ms preprocess, 1.6ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/nhai_training/night_vision_model
[✓] Training Complete! Ready for NHA! deployment.
```

5. Scalability and Economic Impact

5.1 Infinite Data Scalability

Because the foundational data pipeline is entirely synthetic, the project is infinitely scalable to any future needs of the NHA!. If the government introduces a new classification of road sign, or if there is a need to train the system for dense fog conditions in Northern India, we do not need to send fleets of vehicles to collect data. We simply adjust the parameters in the Godot engine, generate 10,000 new synthetic images, and retrain the model in a matter of hours.

5.2 Rapid Fleet Deployment

Scaling physical hardware across 146,000 km of road is a logistical nightmare. Synth-Reflect AI sidesteps this by utilizing existing infrastructure. The model can be deployed as a lightweight software update to dashcams already present in NHA! patrol vehicles, commercial trucking fleets, and public transport buses. This crowd-sourced data model transforms every equipped vehicle into an automated highway auditor, exponentially increasing the frequency of safety checks at virtually zero added hardware cost.

6. Sustainability and Human Safety

6.1 Mitigating Risk

The most immediate benefit of Synth-Reflect AI is the preservation of human life. Traditional road audits require maintenance crews to park on the shoulder of high-speed corridors at night to visually

inspect and log signage degradation. By fully automating this process from moving vehicles, we remove workers from the primary zone of danger, drastically reducing occupational hazards.

6.2 Environmental Sustainability

From a sustainability perspective, eliminating the need for dedicated survey fleets to burn fossil fuels strictly for data collection represents a massive reduction in the NHAI's carbon footprint. The entire data generation phase of this project required zero physical fuel consumption, relying entirely on the computational power of a centralized GPU. Furthermore, the high energy efficiency of the YOLOv8 Nano model ensures that the edge-inference requires a negligible power draw from the host vehicle.

7. Technical Implementation Stack

The architecture relies on a strict division between high-power cloud training and low-power edge deployment.

System Phase	Technology Utilized	Primary Function
Data Generation	Godot Engine 4.3	3D simulation of highway environments, headlight physics, and automated bounding box generation.
Model Training Environment	Google Colab (Tesla T4 GPU)	High-speed cloud processing to iterate machine learning epochs over synthetic datasets.
Computer Vision Architecture	Ultralytics YOLOv8n (Nano)	Single-pass neural network optimized for real-time bounding box detection on low-power edge devices.
Deployment Pipeline	Python / OpenCV	Video frame extraction and confidence-threshold filtering for final dashcam inference.

Training Hyperparameters configuration example

```
results = model.train(  
    data='/content/nhai_dataset/data.yaml',  
    epochs=50,  
    imgsz=640,  
    batch=16,  
    optimizer='auto',  
    project='/content/nhai_training',  
    name='night_vision_model_edge'  
)
```

8. Conclusion

Synth-Reflect AI fundamentally redefines how national infrastructure can be maintained. By utilizing synthetic data generation, we have completely removed the massive cost, time, and physical danger associated with acquiring computer vision training data. By deploying a highly efficient YOLOv8 Nano edge model, we ensure that the software can be scaled across India's 146,000+ kilometer highway network utilizing cheap, existing dashcam hardware. This solution directly supports the MoRTH and NHAI goals of increasing road safety and reducing night-time accident fatalities through proactive, data-driven infrastructure management.

9. Demo

Link: [📺 demo video](#)

References

1. Ministry of Road Transport and Highways. (2025). *Year End Review 2024*. Press Information Bureau. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2091508>
2. DD News. (2026). *India's National Highways record 60% growth in last 10 years to become 2nd largest network in world*. <https://ddnews.gov.in/en/indias-national-highways-record-60-growth-in-last-10-years-to-become-2nd-largest-network-in-world/>
3. Transportation Research and Injury Prevention Centre - IIT Delhi. (2023). *ROAD SAFETY IN INDIA STATUS REPORT 2023*. https://tripc.iitd.ac.in/assets/publication/RSI_2023_web.pdf